# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679


## CLASS 10 — SCIENCE · HARD LEVEL UNIT TEST SET 2

Chapters 9–12: **Light - Reflection and Refraction, The Human Eye and the Colourful World, Electricity, Magnetic Effects of Electric Current** | RBSE Pattern | Session 2026–27

**Time: 2 Hours** Name: _____ Roll No.: _____ **Maximum Marks: 50**

**General Instructions:**

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

## SECTION — A (Objective Type Questions) 15 × 1 = 15

*Choose the correct option and write it in your answer book.*

**1.** Assertion (A): A straight current-carrying wire produces a magnetic field around it. Reason (R): A compass needle placed near the wire gets deflected when the current flows. 1

**(a)** Both A and R are true and R is the correct explanation of A.
**(b)** Both A and R are true but R is not the correct explanation of A.
**(c)** A is true but R is false.
**(d)** A is false but R is true.

**2.** A concave mirror forms a virtual image of magnification +2. The object lies at 20 cm. The focal length is 1

**(a)** -20 cm **(b)** -40 cm **(c)** -10 cm **(d)** -80 cm

**3.** A person can see only from 40 cm to 100 cm. The lens power for distance and reading lens power (to read at 25 cm) are respectively 1

**(a)** -2.5 D and +1.5 D **(b)** -1 D and +1.5 D
**(c)** -1 D and +1 D **(d)** -0.4 D and +3 D

**4.** A wire is stretched to three times its length with its volume unchanged. The new resistance is 1

**(a)** R / 3 **(b)** 9 R **(c)** R / 9 **(d)** 3 R

**5.** A coil moves along the field lines of a uniform field in a direction parallel to the field. The induced current is 1

**(a)** half **(b)** zero **(c)** maximum **(d)** alternating

**6.** When the object distance in a convex lens is made 3 times f, the image distance is 1

**(a)** 2 f **(b)** 1.5 f **(c)** 3 f **(d)** 0.75 f

**7.** The sky appears blue because molecules of the atmosphere scatter 1

**(a)** all colours equally **(b)** only yellow light
**(c)** blue light more strongly than red light **(d)** red light more strongly than blue light

**8.** In the circuit, ammeter reads 2 A and voltmeter across the resistor reads 10 V. The resistance is 1

**(a)** 5 ohm **(b)** 20 ohm **(c)** 12 ohm **(d)** 0.2 ohm

**9.** A magnet is brought toward a coil at a speed v and the induced emf is E. If the speed is tripled, the emf becomes 1

**(a)** E **(b)** 3 E **(c)** E / 3 **(d)** 9 E

**10.** Assertion (A): The image formed by a plane mirror is real. Reason (R): Plane mirror forms an image behind the mirror. 1

**(a)** Both A and R are true and R is the correct explanation of A.
**(b)** Both A and R are true but R is not the correct explanation of A.
**(c)** A is true but R is false.
**(d)** A is false but R is true.

**11.** Assertion (A): Violet light is deviated most by a prism. Reason (R): The refractive index of glass is maximum for violet light. 1

**(a)** Both A and R are true and R is the correct explanation of A.
**(b)** Both A and R are true but R is not the correct explanation of A.
**(c)** A is true but R is false.
**(d)** A is false but R is true.

**12.** Assertion (A): The resistance of a conductor increases with temperature. Reason (R): At a high temperature, the electrons collide more often with the atoms. 1

**(a)** Both A and R are true and R is the correct explanation of A.
**(b)** Both A and R are true but R is not the correct explanation of A.
**(c)** A is true but R is false.
**(d)** A is false but R is true.

**13.** **Assertion (A):** Two magnetic field lines never intersect each other. 1
**Reason (R):** At any point in a magnetic field, the field has only one direction.
Choose the correct option:

**(a)** Both A and R are true and R is the correct explanation of A.
**(b)** Both A and R are true but R is not the correct explanation of A.
**(c)** A is true but R is false.
**(d)** A is false but R is true.

**14.** **Assertion (A):** A ray of light passing through the centre of curvature of a concave mirror retraces its path after reflection. 1
**Reason (R):** The radius of curvature of a spherical mirror is twice its focal length.
Choose the correct option:

**(a)** Both A and R are true and R is the correct explanation of A.
**(b)** Both A and R are true but R is not the correct explanation of A.
**(c)** A is true but R is false.
**(d)** A is false but R is true.

**15.** **Assertion (A):** A rainbow is seen when the Sun is behind the observer and rain is falling in front. 1
**Reason (R):** A rainbow is formed by dispersion, internal reflection and refraction of sunlight in water droplets.
Choose the correct option:

**(a)** Both A and R are true and R is the correct explanation of A.
**(b)** Both A and R are true but R is not the correct explanation of A.
**(c)** A is true but R is false.
**(d)** A is false but R is true.

## SECTION — B (Very Short Answer Type) — 5 × 2 = 10

*Answer in about 20–30 words each.*

**16.** If the angle of the incidence on the prism is varied, will the colours separate in the same way? State why. 2

**17.** A heater of 800 W is connected to a 220 V source. Find the resistance and the heat produced in 5 minutes. 2

**18.** Calculate the current drawn by a 2.2 kW geyser at 220 V and say whether a 5 A or a 15 A circuit is suitable. 2

**19.** Find the lateral shift reasoning: why is it zero for normal incidence? 2

**20.** How does the iris protect the eye and how does the lens focus light? 2

## SECTION — C (Short Answer Type) — 5 × 3 = 15

*Answer in about 40–60 words each.*

**21.** Why should an electrical appliance such as a toaster have a metal casing connected to the earth wire? 3

**22.** A ray of light enters from glass (n = 1.5) to water (n = 1.33). Does it bend towards the normal? Find the relative refractive index. 3

**23.** An object is placed at 10 cm and 100 cm in front of a normal eye. At which position does the ciliary muscle contract more and why? 3

**24.** Why are ammeters of low resistance and voltmeters of high resistance? 3

OR

Why do we say that electricity and magnetism are related? Give two examples. 3

**25.** A convex lens forms a real image of a candle on a screen 80 cm away. The magnification is -3. Find u and f. 3

OR

An elderly person has a near point of 1 m and a far point of 2 m. Describe the defects and the lens prescription for reading at 25 cm and seeing far. 3

## SECTION — D (Long Answer Type) — 2 × 5 = 10

*Answer in about 100–150 words each.*

**26.** **Case study.** Ravi, a Class 10 student, is not able to read the writing on the blackboard from the last bench, though he reads his book comfortably. An eye specialist finds that the farthest point at which he can see clearly is 80 cm from his eye. 5

(i) Name the defect of vision from which Ravi suffers.
(ii) What type of lens will correct this defect?
(iii) Give one cause of this defect.
(iv) Calculate the power of the lens required to correct it.

**27.** An electric heater has two coils of 20 ohm and 30 ohm that can be used singly, in series or in parallel on a 120 V supply. Find the power in each of the four modes and identify the highest-power mode. 5

OR

Compare the magnetic field of a bar magnet and of a current-carrying solenoid. Explain how a solenoid can be used to lift iron scrap. 5

## END OF QUESTION PAPER — Total: 50 Marks